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WTP - 2

STP

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Utility Section

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Sr.	Equipment Name	Application	Make	Frame Size	Foot or Range Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	Direct Mount/ Pulley/ Gear	VFD Installed, V/FV Drive, Hz & Make	Daily Running Hours	Annual Operating Days/ Year	Observations	Measured Voltage	Measured Current	Measured KW	Measured PF
25	4878 Chiller	Production	Blue Star	80								24	365	Power factor is 0.75	420	37.3	21.5	0.75
															415.6	37.3	20.12	0.739
															414.9	37.3		
															416.4	35.5	21.75	0.763
26	44 T/C Chiller	Production	Blue Star	80								24	365	Power factor is 0.75	420	37.3	21.5	0.75
															415.6	37.3	20.12	0.739
															414.9	37.3		
															416.4	35.5	21.75	0.763
26	34 T/C Production	Production										24	365		414.9	61.3	33.87	0.765
															411.9	64.8	36.83	0.732
															416.4	61.2	36.15	0.814
															411.9	64.8	36.83	0.732
26	34 T/C Production	Production										24	365		414.9	61.3	33.87	0.765
															411.9	64.8	36.83	0.732
															416.4	61.2	36.15	0.814
															411.9	64.8	36.83	0.732
26	44 T/C Pump														422.8	6.77	4.518	0.910
															421.7	6.49	4.182	0.982
															416.4	61.2	36.15	0.814
															411.9	64.8	36.83	0.732

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Unit 1 (Production A)

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF	
1	50 50 50	Supply Air	10500 CFM	140					11				141 141 141	42	353	108	2700V	811	1662	1662	0.81
2	50	Return Air	10500 CFM	65					5.5				101 101 101	42	353	108	4150V	11.5	855	855	0.85
3	70 70 70	Supply Air	6500 CFM	100					5.5				490 490 490	42	353	108	4325V	23.3	159	159	0.83
4	70	Return Air	6500 CFM	65					2.5				490 490 490	42	353	108	4325V	0.8	1662	1662	0.81

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Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
	PA1 04	Supply Air	7500	100					5.5				49 50Hz Danfoss	24	353	Belt	416 439.1	4.34 7.48	3.106 3.945	0.949 0.798
	PA1 12	Supply Air	8000	50					7.5				49 40Hz Danfoss	24	353	Belt	416 428	4.34 5.65	3.106 2.825	0.949 0.996
	PA1 13	Supply Air	8000	100					5.5				49 42.5 Hz Danfoss	24	353	Belt	416 402.8	4.34 4.87	3.106 1.958	0.949 0.706

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
		Return Air	5000	65					9.7 100				409 112	24	353	Belt	R 404.3 Y 402.9 B 405.6 Avg	5.60 10.4 6.59	1.735 4.565 2.430	0.698 0.674 0.709
		Heater							21 100							TPH320	R 444.7 Y 437.7 B Avg	2.07 1.98	1.326 1.143	0.832 0.863
	PA AHU 11	Supply Air	1600	100					1.5 100				409 48.7	24	353	Belt	R 444.7 Y 437.7 B Avg	2.07 1.98	1.326 1.143	0.832 0.863
		Return Air	1600	65					1.1 100				409 29.7	24	353	Belt	R 444.7 Y 437.7 B Avg	2.07 1.98	1.326 1.143	0.832 0.863
		Heater							2 100							TPH320	R 444.7 Y 437.7 B Avg	2.07 1.98	1.326 1.143	0.832 0.863

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
	PA AHU 10	Supply Air	1500	100					1.5				485 V 24 Pan	24	353	Belt	R 484 Y 484.2	1.57 0.914 1.62	0.149 0.103 0.768	0.830 0.754
		Return Air	1500	65					0.75 20				409 V 24 Pan	24	353	Belt	R 402.3 Y 398.8	3.42 3.15 1.978	0.914 0.908	
		Weather							3 20							422 7.20 4.419	0.998			
	PA AHU 02	Supply Air	6600	100					11 20				420.8 V 24 Pan	24	353	Belt	R 420.8 Y 420.8	8.16 5.010	0.983	
		Return Air	6600	65					3.7 20				416.9 V 24 Pan	24	353	Belt	R 416.9 Y 416.9	1.01 0.512	0.997	

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven		Measured Voltage	Measured Current	Measured KW	Measured PF
	Pt / AHU / 06	Supply Air	6400	140					9.3				49.5 Hz Dan			Belt	R	423	11.2	7.050	0.996
		Return Air	6400	65					3.7				98.9 Hz Dan			Belt	Y	423	12	7.929	0.992
									100								B				
																	Aug				
		Heater							28.5							TH mixer	R	402.8	3.6	24.79	0.779
									100								Y				
																	B				
																	Aug				
	Pt / AHU / 5	Supply Air	8000	100					18.5				49 Hz Dan			Belt	R	403.7	26.8	14.59	0.911
									100								Y				
																	B				
																	Aug				
		Return Air	8000	65					3.7				49 Hz Dan			Belt	R	420.8	2.92	1.808	0.995
									100								Y	421.2	2.89	1.888	0.997
																	B				
																	Aug				

Sr.	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
1	Heater	1500						12							(DFT)				
2	PT / AHU	100						11				38 Hz			belt	421.5	7.23	4.0099	0.978
3	PT / AHU	65						8.7				44.8 Hz			belt	418.9	1.38	0.992	1.000

Chiller 2 / U179

Sr.	Equipment Name	Application	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	Direct Mount/ Pulley/Gear	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/ Year	Observations		Measured Voltage	Measured Current	Measured KW	Measured PF
16	chiller 2 (2007R)	✓	Not			130.9					No	24	365						
	U179																		
	CT for					7.5 hp					No	24	365						
	cooling tower pump - 2 (compressor pump)				B3	380 hp					No	24	365						
	water circulation pump - 2										No	24	365						

$$S = \frac{125 \text{ m}^3/\text{hr}}{20 \text{ m}} = 2.4 \text{ kg/s}$$

$$d = 4.5 \text{ kg}$$

Sr.	Equipment Name	Application	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	Direct Mount/ Pulley/Gear	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/ Year	Observations	Measured Voltage	Measured Current	Measured KW	Measured PF
	Chiller-3 190 FD		Yarc								NO	24	365	Ch3 1228.0				
	CG FED 250 271				Flange 150				570 RPM		NO	24	365					
	COOLING TOWER Circulation pump -2				885	18.5 FED		IE3	1480		NO	24	365	190 m3/hr 25m				
	COOLING Circulation pump -2				885	18.5 FED		IE3	1480		NO	24	365	NO 25m 5.8 m3/hr Discharge				

ATCS Not working

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Sr.	Equipment Name	Application	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	Direct Mount/ Pulley/Gear	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/ Year	Observations		Measured Voltage	Measured Current	Measured KW	Measured PF
	Air compressor Condenser					5.5 1000						24	365		Avg R Y B	414.9 424.4 420.9	7.04 6.88 7.16	2.565 2.575 2.715	0.495 0.505 0.512
	Air compressor Cooling to water pump					7.5									Avg R Y B	419.8 420 428.4	6.72 6.63 5.90	2.845 2.889 3.458	0.799 0.796 0.778
	Air compressor 42.6 CPM					15 1000					Soft start 24	365		6.8 for Nedra	Avg R Y B	424.7 420.5 421.6	8.68 8.91 8.01	5.903 5.764 5.753	0.904 0.887 0.882
	Air dryer		parted			5.5 1000		81%			24	365	6.8		Avg R Y B	416.9 417.7 416.7	10.8 10.9 10.7	5.368 5.426 5.318	0.684 0.678 0.687
	Water pump No. 2					2.5 1000					24	365			Avg R Y B	416.9 417.7 416.7	10.8 10.9 10.7	5.368 5.426 5.318	0.684 0.678 0.687

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Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
48	PC FDV 01	Supply AIR	1500	60					0.75 100				20	24	353	Belt	403.8 406.1	2.07 2.18	1.101 1.312	0.702 0.762
		Return AIR	1500	50					0.5 100				20	24	353	Belt	Both Buses from			
	PB AHU 04	Supply AIR	1184						0.3 100				433 112 Danco	24	353	Belt	403.1 402.2	12 11.35	6.452 5.843	0.985 1.000
	PB AHU 03	Supply AIR	1065.7						11 100				409 95 112 Danco	24	353	Belt			1.95	
	PB AHU 02	Supply AIR	1057.5						15 100				409 51.5 112	24	353	Belt	401.9 406.9	24.8 24.5	16.61 16.54	0.999 0.946

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Hrs & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/ Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF	
	$\frac{P3}{AHU 01}$	supply Air	6500 CFM						92 KW				Y49 ABB 38 V12	24	353	Belt	R	427.2	8.91	4.729	0.999
																	Y	427.5	10.2	5.492	0.955
																	B	422.4	11.3	6.319	0.999
		Speaker							48 KW							THY 1100V	Avg				
																	R	405	16.6	8.765	0.899
																	Y	402.1	11.8	5.188	0.990
	$\frac{P4}{AHU 01}$	supply Air	6500 CFM						5.5 KW				Y49 Danfoss 95 H12	24	353	Belt	Avg				
																	R	424.9	8.5	1.916	0.998
																	Y	425.6	2.85	1.624	0.998
		Return Air							9.7 KW				Y49 HC Danfoss	24	353	Belt	Avg				
																	R	420.3	1.29	0.689	0.986
																	Y	424.6	1.33	7.043	0.982
	$\frac{P5}{AHU 04}$	supply Air	7000 CFM						7.9 KW				Y49 424.24 J Danfoss	24	353	Belt	Avg				
																	R	412.1	10.1	5.789	0.769
																	Y	412.8	6.885	6.505	0.802
																	B		11.2		
																	Avg				

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
		Return Air	2000 6458	65 140					5.5 100				Yes 35 Hz 1000 -pays	24	353	Belt	R 427.4 B 428.5 Avg	R 8.06 B 8.02 Avg	R 1.672 B 1.658 Avg	R 0.998 B 0.999 Avg
	PC AHU 03	Supply Air	10500	140					15 100				Yes Belt 1000 -pays	24	353	Belt	R 424.9 B 420.1 Avg	R 19.6 B 16.6 Avg	R 7.868 B 8.551 Avg	R 0.753 B 0.722 Avg
		Return Air	11857	65 mm					7.5 100				Yes Belt 1000 -pays	24	353	Belt	R 414.2 B 415.8 Avg	R 8.28 B 8.90 Avg	R 4.969 B 4.389 Avg	R 0.828 B 0.698 Avg
		Heater							20.5 100							thumper (0.45)	R 405.8 B 421 Avg	R 3.53 B 3.61 Avg	R 2.048 B 2.094 Avg	R 0.997 B 1.000 Avg
	PC AHU 02	Supply Air	3630 CFM						5.5 100				Yes VFD 32 Hz 1000 -pays	24	353	Belt	R 405.8 B 421 Avg	R 3.53 B 3.61 Avg	R 2.048 B 2.094 Avg	R 0.997 B 1.000 Avg

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
	$\frac{ST}{AHU}$ 01	Supply Air	1500	140					2.2				49 Dan A01 499 Hz	24	353	Belt	R 422.2 Y 417.2	1.97 1.91	1.325 1.195	1.000 0.999
		Return Air	1077	65					1.1				49 Dan A01 499 Hz	24	353	Belt	R 422.2 Y 417.2	0.83 0.83	0.188 0.188	0.908 0.908
	$\frac{ST}{AHU}$ 02	Supply Air	1500	140					2.2				50 Dan A01 499 Hz	24	353	Belt	R 424.1 Y 424.2	2.26 2.24	1.497 1.449	1.000 0.999
		Return Air	1077	65					1.1				50 Dan A01 499 Hz	24	353	Belt	R 422.2 Y 417.2	1.05 1.05	0.894 0.894	0.860 0.860
	$\frac{ST}{AHU}$ 03	Supply Air	3000						3.7				N	24	353	Belt	R 422.1 Y 417.2	8.06 5.192	5.192 0.882	0.882 0.882
		Return Air	1077	65					1.1				N	24	353	Belt	R 422.1 Y 417.2	8.06 5.192	5.192 0.882	0.882 0.882

Sr.	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
1	Supply Air	800 CFM	125 mm	800 mm				2.2				No	24	353	Belt	422.3	1.71	1.222	0.919
2	Return Air	800 CFM	75 mm	800 mm				0.75				No	24	353	Belt	427.2	0.86	0.484	0.7929
3	Supply Air	1700 CFM	100 mm	800 mm				15				Yes	24	353	Belt	425.8	10.2	6.612	0.913
4	Supply Air	1700 CFM	100 mm	800 mm				15				Yes	24	353	Belt	412.1	5.03	2.901	0.86
5	Supply Air	1700 CFM	100 mm	800 mm				15				Yes	24	353	Belt	407.8	5.10	2.953	0.722
6	Supply Air	1700 CFM	100 mm	800 mm				15				Yes	24	353	Belt	406.8	2.37	1.565	0.944
7	Supply Air	1700 CFM	100 mm	800 mm				15				Yes	24	353	Belt	423.7	2.42	1.563	0.944

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD installed, (Y/N), Oper Hrs & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
	B AHU 07	Supply Air	8000	100					7.5				Schneider 40 HL	24	353	Belt	R 404.3	3.37	2.095	0.990
		Return Air							5.5				SGH2 40 HL Schneider	29	353	Belt	Y 407.3	3.28	2.179	0.999
	B AHU 13	Supply Air	1500 CFM	100					1.5				Yes 50 Schneider	29	353	Belt	R 407.8	5.10	2.588	0.799
		Return Air		65					1.1				No	24	353	Belt	Y 410.1	5.03	2.591	0.802
																	B 411.1	1.41	0.991	0.985
																	Avg 406.9	1.92	2.397	0.975
	B AHU 05	Supply Air	6000	100					5.5				Yes 45 HL	24	353	Belt	R 410	2.49	1.708	0.990
																	Y 408	2.80	1.982	0.995
																	B			
																	Avg			

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Hz & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
	$\frac{B}{AHU} \frac{14}{14}$	Supply Air	1500	100					1.5 KW				Yes 40Hz Schneider	24	353	Belt	R 407.7 Y 410 Avg	1.71 1.6	1.027 1.005	0.983 0.995
	$\frac{B}{AHU}$	Return Air		65					1.1 KW				Yes 26 Hz Schneider	24	353	Belt	R 402.2 Y 406.4 Avg	0.45 0.42	0.257 0.23	0.914 0.891
	$\frac{B}{AHU} \frac{10}{10}$	Supply Air	600	100					5.5 KW				Yes 24 Hz Dantex	24	353	Belt	R 404.5 Y 405.6 Avg	3.58 2.54	1.308 1.209	0.977 0.991
		Return Air		85					5.2 KW				Yes 27 Hz Schneider	24	353	Belt	R 404.5 Y 400.2 Avg	1.59 1.57	1.004 0.984	0.919 0.909

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, INWG	Measured Static Pressure, INWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VD Installed, V/N, Oper Hz & Make	Daily Running Hours	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
	B AHU 06	Supply Air	5000	100					37				4y Scheidt 90H2	24	353	belt	R 403.8	2.67	1.810	0.992
																	Y 400.5	2.72	1.742	0.988
																	B			
																	Avg			
		Return Air		65					37				4y Scheidt SDH2	24	353	belt	R 403.5	3.15	2.143	0.991
																	Y 399.7	3.38	2.208	0.992
																	B			
																	Avg			
		Heater				Hydram			9 FW							THYRISTO	R 403.8	9.93	6.496	0.972
																	Y 398.6	10	6.279	0.945
																	B			
																	Avg			
76	B FDV V5-02		5000	75					37				NA DOL	24	353		R 402.4	5.17	2.762	0.813
																	Y 401.8	5.24	2.784	0.748
																	B			
																	Avg			
78	B FDV NR-02		7000	75					5.5				NA DOL	24	353		R 403.6	6.63	4.110	0.870
																	Y 398.6	6.98	3.930	0.83
																	B			
																	Avg			

Sr.	AHU Fan	Application	Design Flow, CFM	Static Pressure, MMWG	Measured Static Pressure, MMWG	Make	Frame Size	Foot or Flange Mounted	Rated Power, KW	Rated Efficiency, %	IE Class (IE1/IE2/IE3)	Rated RPM	VFD Installed, (Y/N), Oper Ht. & Make	Daily Running Hours Hrs/Day	Annual Operating Days, Days/Year	belt / direct driven	Measured Voltage	Measured Current	Measured KW	Measured PF
	B AHU 12	Supply Air	2000						2.2 KW				Bypass	24	353		R 406.5	2.46	1.307	0.789
																	Y 402.8	2.38	1.449	0.756
																	B			
																	Avg			
		Return Air							1.5 KW				Rel Schneider	24	353		R 408.7	0.59	0.379	0.945
													34 Hz				Y 409.3	0.59	0.368	0.959
																	B			
																	Avg			
	B AHU 04	Supply Air	10000						11 KW				Rel Schneider	24	353		R 414.3	7.89	5.081	0.993
													47 Hz				Y 410.8	7.18	4.933	0.995
																	B			
																	Avg			
		Return Air							5.5 KW				Rel Schneider	24	353		R 410.1	5.88	3.498	0.798
													27 Hz				Y 411.8	5.76	3.790	0.849
																	B			
																	Avg			
																	R			
																	Y			
																	B			
																	Avg			